

A Graph ✓ valid

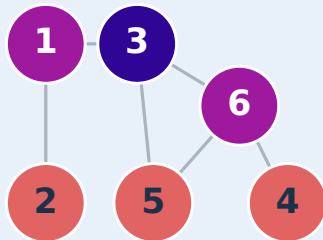
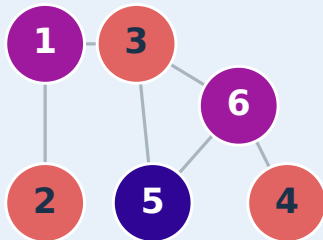
partial colors 

fill vertices 2, 3, 5

1 2 3 5



2 2 3 5



B Countdown ✓ = 18

tiles {3, 6, 9}

target 18

use each tile once

1 $(6 \times 9) / 3$

`div(mul(6,9),3)`

2 $6 + 3 + 9$

`add(9,add(3,6))`

C Python ✓ divides

lambda n: EXPR

one call per n in

18, 82, 91, 93

1 $2 \text{ if } n\%2==0 \text{ else } 7 \text{ if } n\%7==0 \text{ else } 3$

`[2, 2, 7, 3]`

2 $3 \text{ if } n\%3==0 \text{ else } 13 \text{ if } n\%13==0 \text{ else } 41$

`[3, 41, 13, 3]`

D MathIR ✓ $x = 16$

solve $x/2 - 9 = -1$

... · C: add 9 · D: add 1

E: add 18 · F: $\times 2$

1 C;F

add 9, then $\times 2$
 $x/2 = 8 \rightarrow x = 16$

2 F;E

$\times 2$, then add 18
 $x - 18 = -2 \rightarrow x = 16$

E PantryPlan ✓ feasible

2-4 ingredients

125-200 g total, four bounds

reply: six row bits

1 000011



2 100100

